







Trinity College Dublin
Commodities Society

Outage-Driven Power Price Volatility in the I-SEM:

A Technology Response Perspective

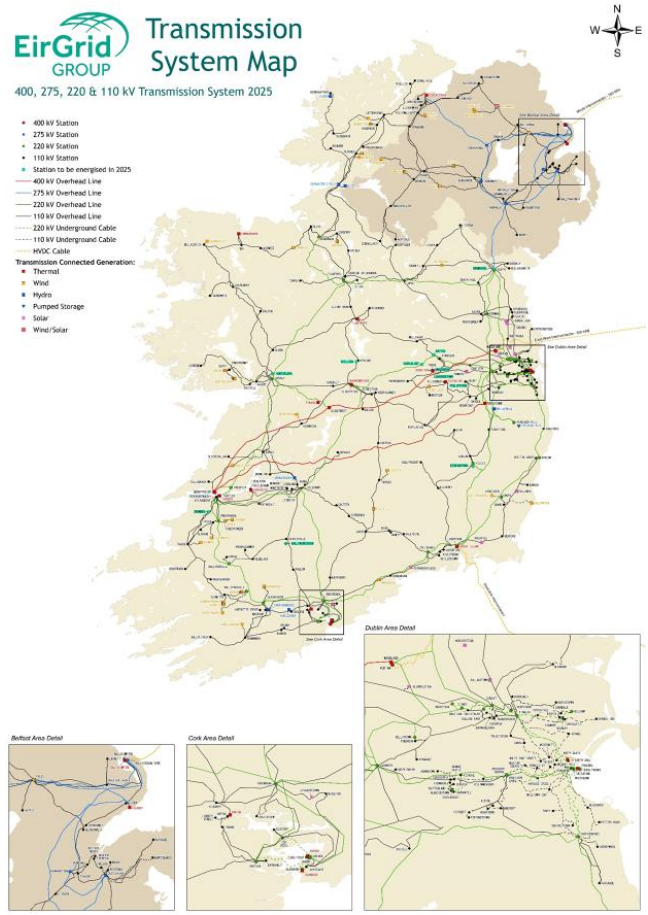
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Introduction

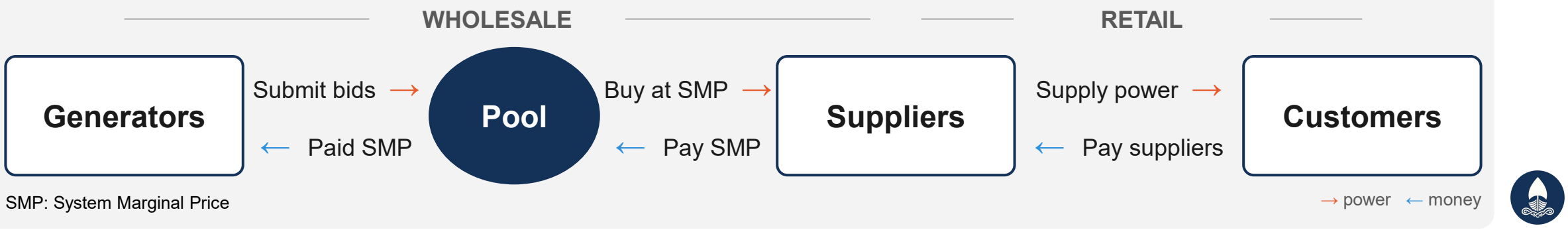
- Power is traded like any commodity, but balanced **second-by-second**
- Power markets are an **economic reflection of the physical grid**
- More renewables + less synchronous generation = **more volatility** [1]

This presentation looks at:

1. How power is traded in Ireland
2. How the grid and markets influence each other
3. An extreme case: when things go wrong
4. Technologies that can stabilise grid security and prices



Source: EirGrid [2]



A Current Overview of Ireland's Grid

Approx. 1/3 of electricity from wind in 2025, one of the highest shares in the world [3]

3 HVDC links to GB (EWIC, Moyle, Greenlink); 700 MW Celtic link to France under construction [4]

Target: 80% renewable electricity by 2030 [5]

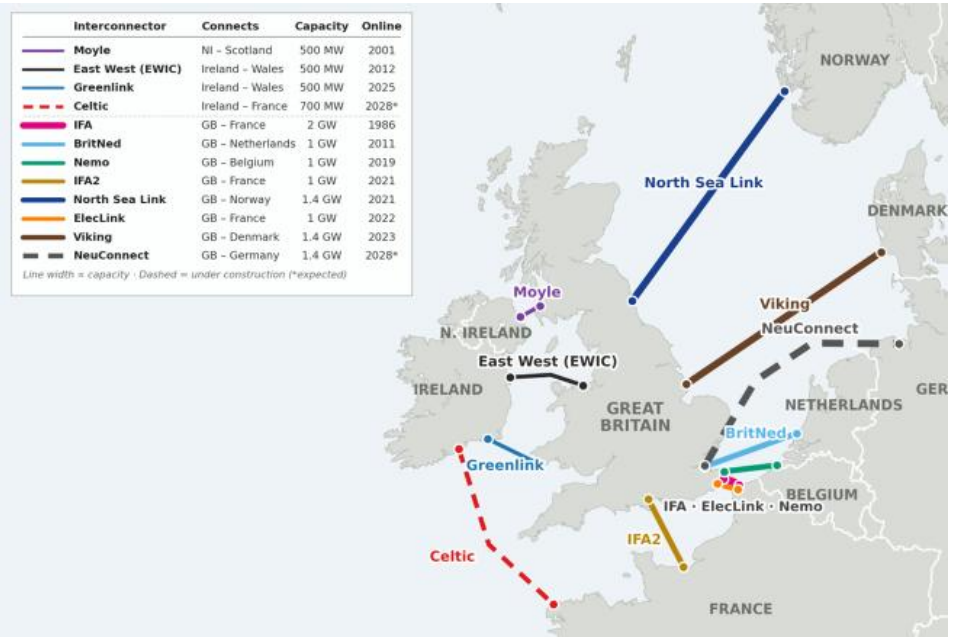
The issue: wind and solar are non-synchronous. They provide no inertia, and output can collapse in still weather. [1]

All Island (I-SEM) 2026 YTD uncurtailed statistics (to 29 Sep)

Quantity (MW)	Mean	P10	P90	Max	Min
Demand	4818	3939	5872	7339	3371
Wind generation	1575	318	3144	4713	0
Daily solar peak	588	–	–	1402	36

Source: Private database [6]

Technology	Today	By 2030
Wind	5 GW [3]	9 GW on + 5 GW offshore [5]
Solar	2.3 GW [7]	8 GW [5]
BESS	~1 GW [8], [9]	~2 GW [9]
Thermal	6–6.5 GW [10]	~6 GW [10]
Interconnect	1.5 GW [4]	2.2 GW [4]



Interconnectors: Ireland, GB and Europe [11]

An Introduction to Power Trading



How Is Electricity Traded in Ireland?

- **I-SEM (Integrated Single Electricity Market):** The ROI and NI wholesale electricity market where generators sell and suppliers buy. [12]

Closer to delivery

DAM

- Main auction, closes at 11:00 the day before delivery, holds the most liquidity.

IDA1–3

- Smaller auctions to adjust positions closer to delivery as forecasts and scheduled generation change.

IDC

- Continuous trading until 1 hour before delivery, allowing traders to tighten positions.

Ex-ante trading ↑

Real-time balancing ↓

Balancing Market (BM)

- At gate closure units submit their **Final Physical Notifications** (FPNs), which is their intended output, and prices to increase or decrease their generation (**incs** and **decs**).
- The **TSOs** (EirGrid and SONI) accept incs/decs to **match supply and demand**, keeping the system balanced.
- An **Imbalance Price** is derived from each 5-minute balancing period and averaged over each 30-min settlement period. [12]
- Depending on the system **Net Imbalance Volume** (NIV), generators either buy back or sell power to the grid at this price.

GB-Coupled (IDA1 & 2)

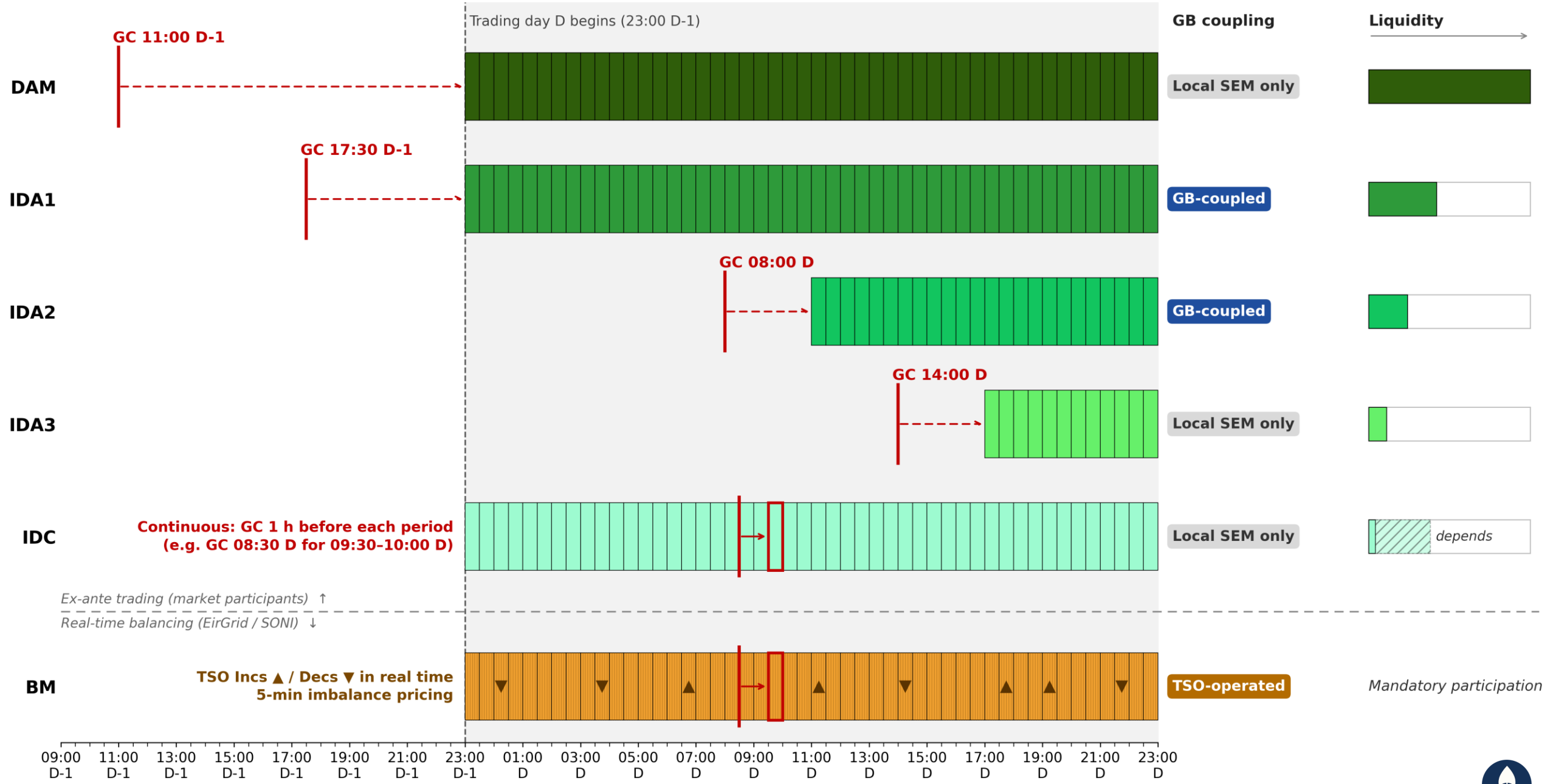
Irish and GB IDA1 & 2 bids clear together, and interconnector flows follow the price difference

The Trade-Off

Trading Early: Gives traders price certainty by locking in prices early, but uses older forecast runs.

Trading Later: Better forecasts, but thinner liquidity to buy and sell volumes.

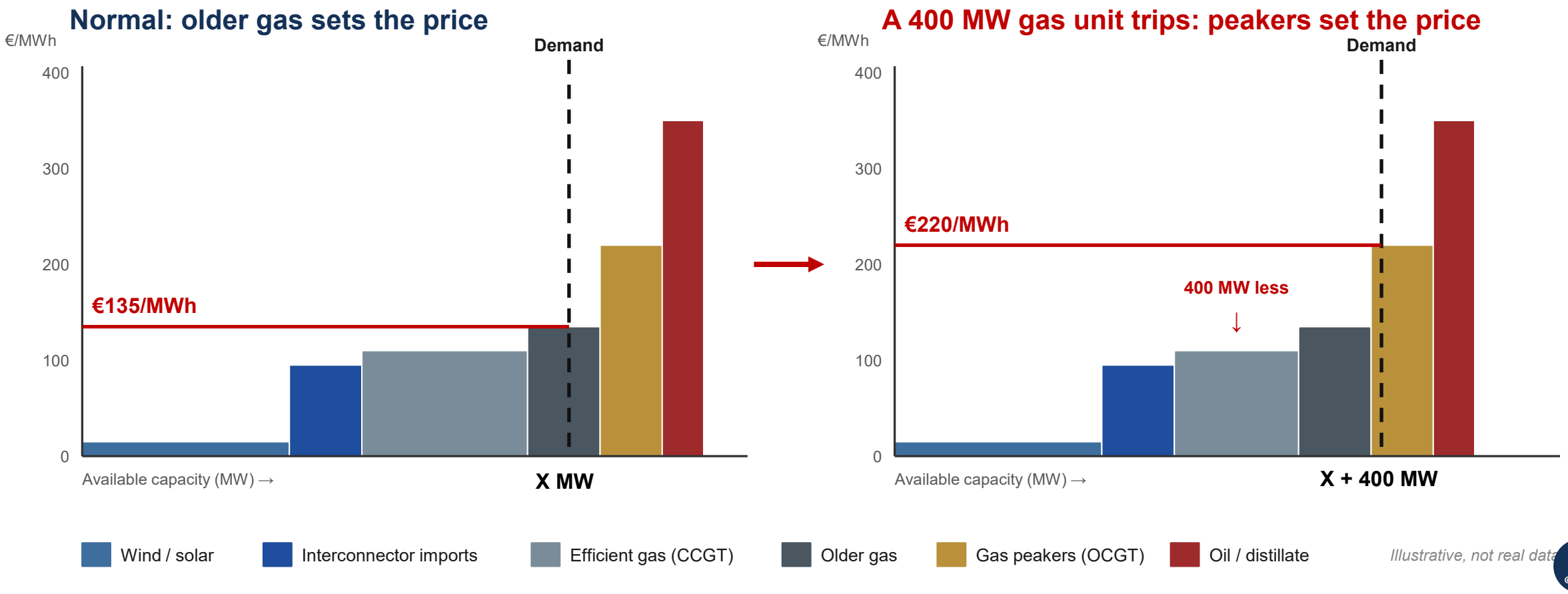
I-SEM Market Timeline: From Day-Ahead Trading to Real-Time Balancing



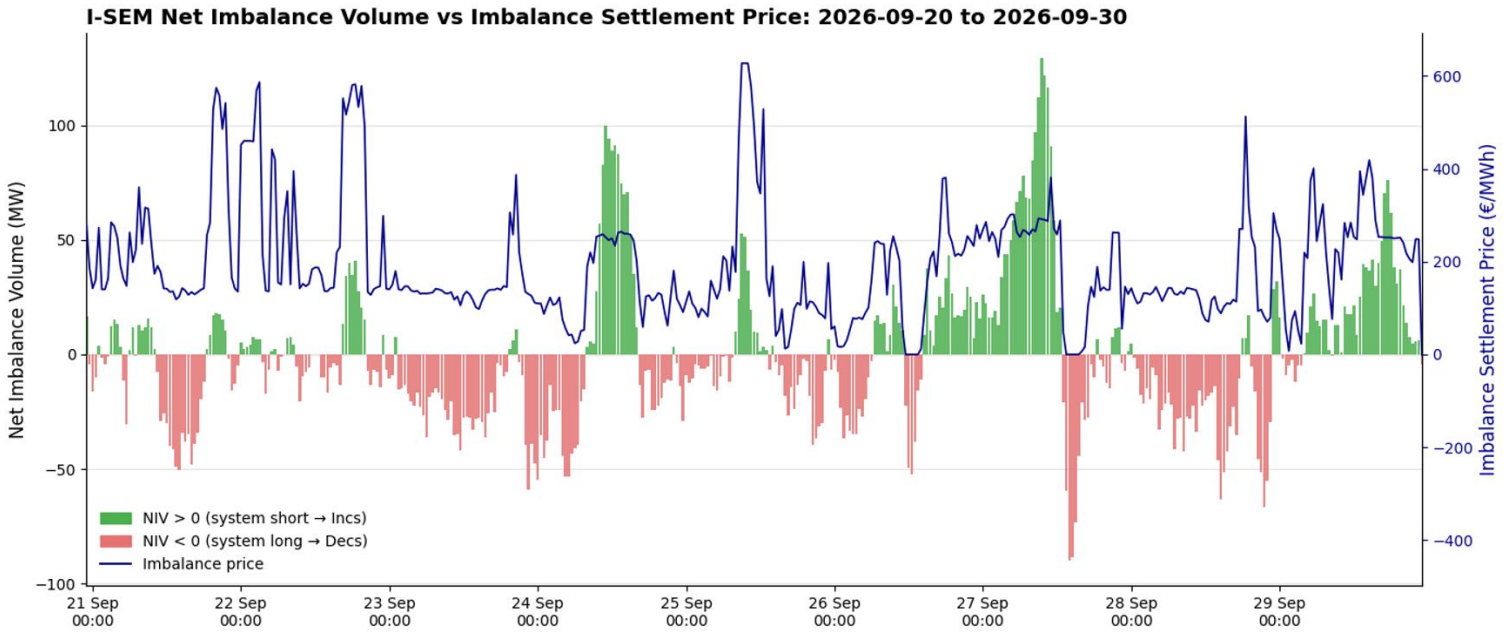
Times in Irish local time. Red line = gate closure; arrow = lead time to delivery. GB-coupled auctions trade via the Moyle and EWIC interconnectors. Liquidity bars are an illustrative ranking, not to scale.

How Power Is Priced: Marginal Pricing

1. Generators offer power at a price for each trading period in each auction, stacked cheapest first: the merit order
2. The last unit needed to meet demand sets the price for all participating units
3. In times of low renewable output, Ireland's stack jumps from cheap wind to expensive gas, with little in between. [13]



The Balancing Market: Where Volatility Lands



Graph produced using Python and historic market data [6]

Imbalance statistics, 2026 to date (30/09/26)

[6]

	NIV (MW)	ISP (€/MWh)
Mean	-2.4	€138.7
Max	+129.2	**€627.3**
Min	-238.0	**-€46.3**

≈ €670/MWh swing in the imbalance price this year

One price for all units out of balance

	Unit long	Unit short
System short (high price)	▲ Sells at a high price	▼ Buys back at a high price
System long (low price)	▼ Sells at a low price	▲ Buys back at a low price

▲ System short
TSO incs units up → price spikes

▼ System long
TSO decs units down → price falls, even below zero

€567m
imperfections costs 2024/25; ~€700m forecast 25/26 [14], [15]

The Takeaway: More renewables are making the balancing market more volatile, and those costs flow through to household bills. The grid system needs more flexible capacity that can ramp up and down whenever the TSO calls for it.

Flagging & Tagging: Why Not Every Action Sets the Price

GB and the I-SEM both screen out non-marginal balancing actions before calculating price – just with different criteria

Every balancing action is a **Bid-Offer Acceptance (BOA)**. Not all of them reflect genuine national scarcity – flagging and tagging decide which ones count toward the published price.

GB: Flagging (1 step)

- **Unflagged** = a genuine, system-wide balancing action
- **Flagged** = accepted for another reason – a local constraint, a reserve/response contract, testing
- **System Price** = volume-weighted cost of the unflagged BOAs needed to cover the Net Imbalance Volume [16]
- One question: was this action taken to balance the whole system?

I-SEM: Tagging, then Flagging (2 steps)

- **Tagging**: Energy vs Non-Energy action – Non-Energy (voltage control, stability, local constraints) is excluded immediately
- **Flagging**: among Energy actions, is the unit at a physical limit (min/max generation, ramp rate)? If so, it's excluded as non-marginal
- **Imbalance Price** = marginal 120 MW of flagged-in action per 5-min period, averaged over the half-hour [17]

Same vocabulary, different filter. GB asks why an action was taken. Ireland asks why (tagging) and how (flagging) before a BOA can set the price.

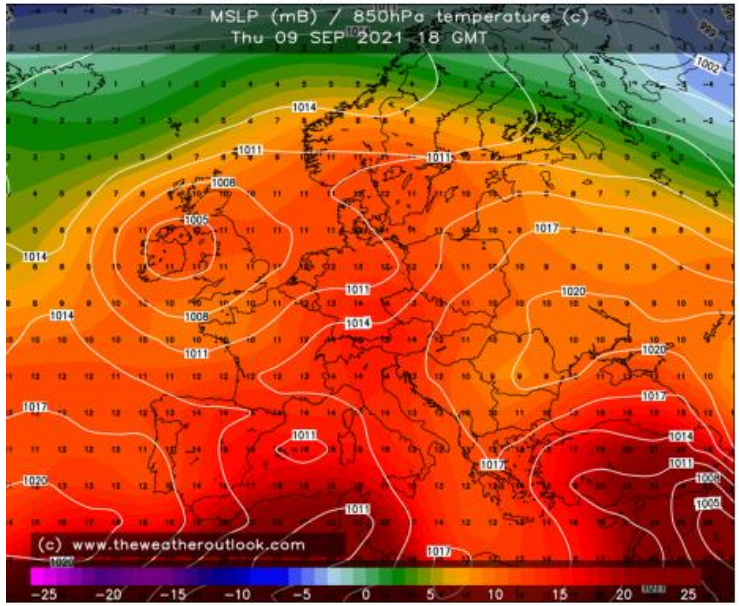


Case Study: Record I-SEM and GB Imbalance Prices

September 9th 2021: Extreme Balancing Market Prices

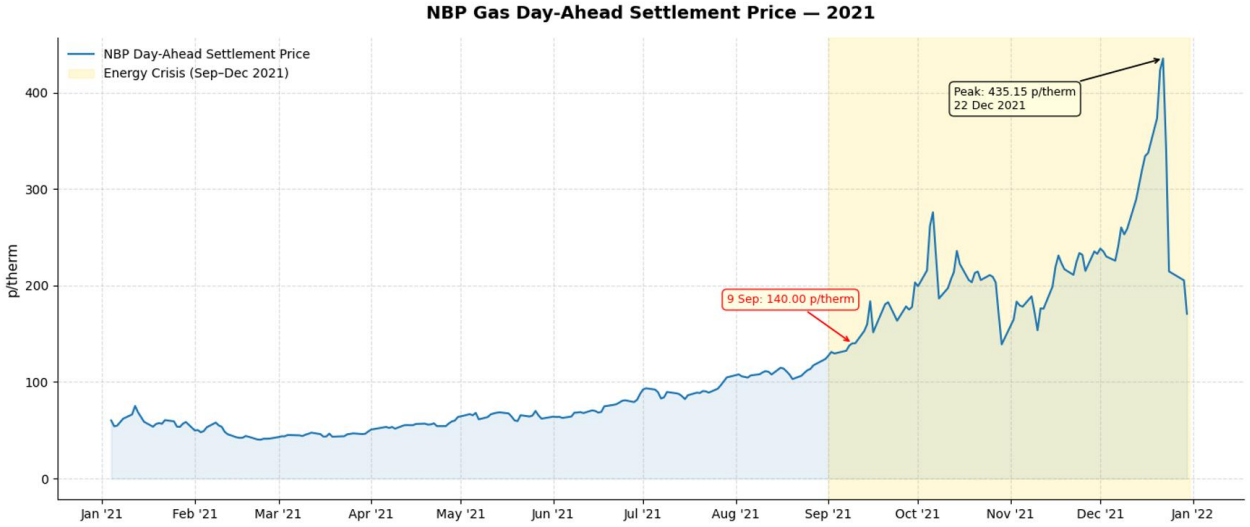
Trading periods 33–38 (16:00–19:00) saw record imbalance prices in both the GB and I-SEM balancing markets

- High-pressure system over the UK & Ireland → very low wind generation in both countries [18]
- Ireland and GB had 2.8+ and 17+ GW of coal, gas, biomass & nuclear offline for summer maintenance [19], [20]
- Landed during the early 2021/22 Russia-Ukraine gas crisis, which was already pushing prices sharply higher
- Result: both systems were extremely tight - some flexible plants opted to not trade into the day ahead or intraday markets, betting on higher balancing prices



140 (€56.35/MWh) → 435 (€175.12/MWh) p/therm

NBP day-ahead gas price on 9 Sep 2021, versus the eventual annual peak on 22 Dec 2021 [6]



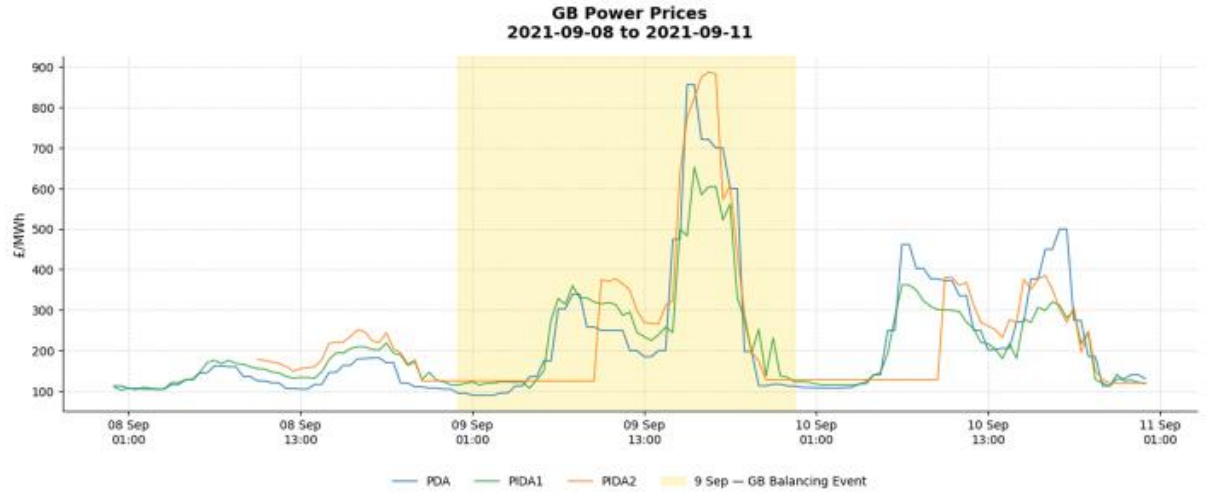
The GB Side

National Grid's (GB TSO) Balancing Mechanism during the 9th of September price spike

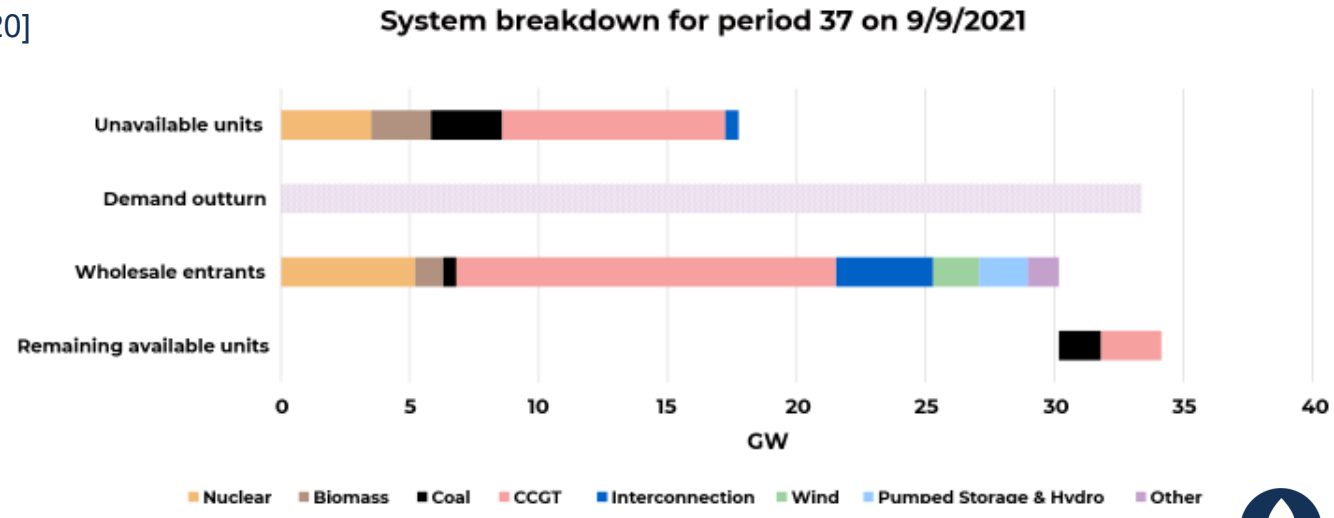
- Periods 33–38 saw GB system prices peak at **£4,037/MWh** [6]
- Demand exceeded **33 GW**, while the cumulative FPN in the wholesale market covered only **~30 GW**; leaving **~3 GW** for the Balancing Mechanism to fill [20]
- With **17+ GW** of capacity on outage, National Grid had just **3.9 GW** of available assets to cover that gap [20]
- Many asset owners anticipated the shortfall, expecting to be instructed to increase output (an "inc") regardless [20]
- Despite wholesale prices being very high, some units strategically withheld capacity, and successfully attempted to utilise the imbalance price cap of £4,000/MWh [21]

3.9 GW

available flexible capacity left to cover a 3 GW deficit, after 17+ GW of outages



GB day-ahead/intraday price series, 8-11 Sep 2021 [6]



GB period 37 system status [20]

The GB Side

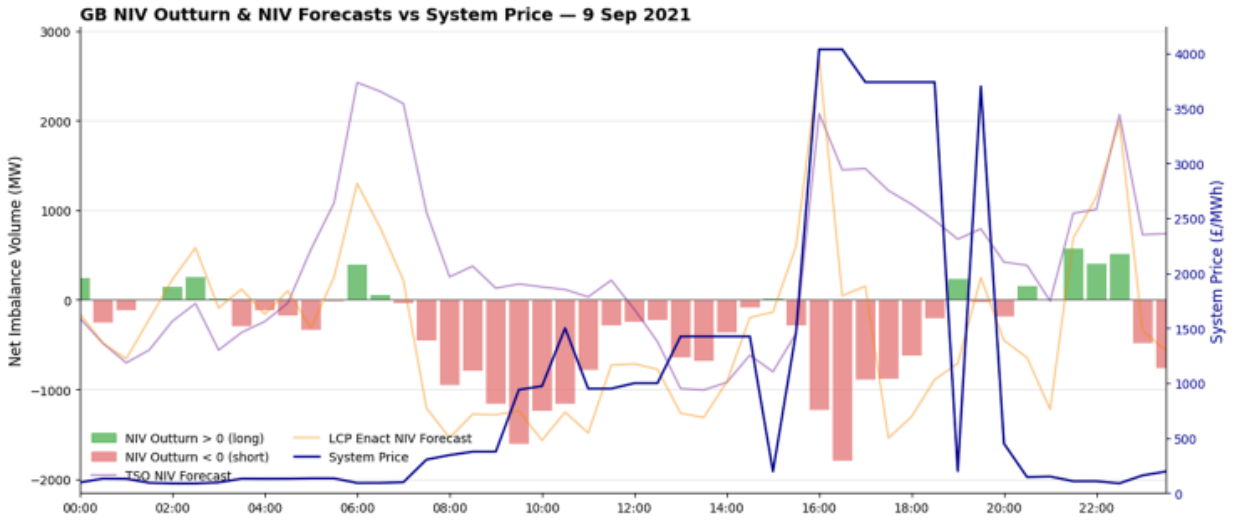
- TSO and third-party forecasts all predicted a long system – the outturn was short, worsening the shortage [21]
- Battery, hydro, CCGT and coal units were dec'd (paid to turn down) – most likely due to local grid congestion, not a nationwide signal [21]
- In period 37, the **highest offers** came from West Burton 1 & 2 – EDF's **coal power plants**, a combined **515 MW at £4,000/MWh** [21]
- These offers were left unflagged by National Grid, so they were used to set the £3,738.8/MWh system price for that period [21]

Bids/Offers Accepted - GB

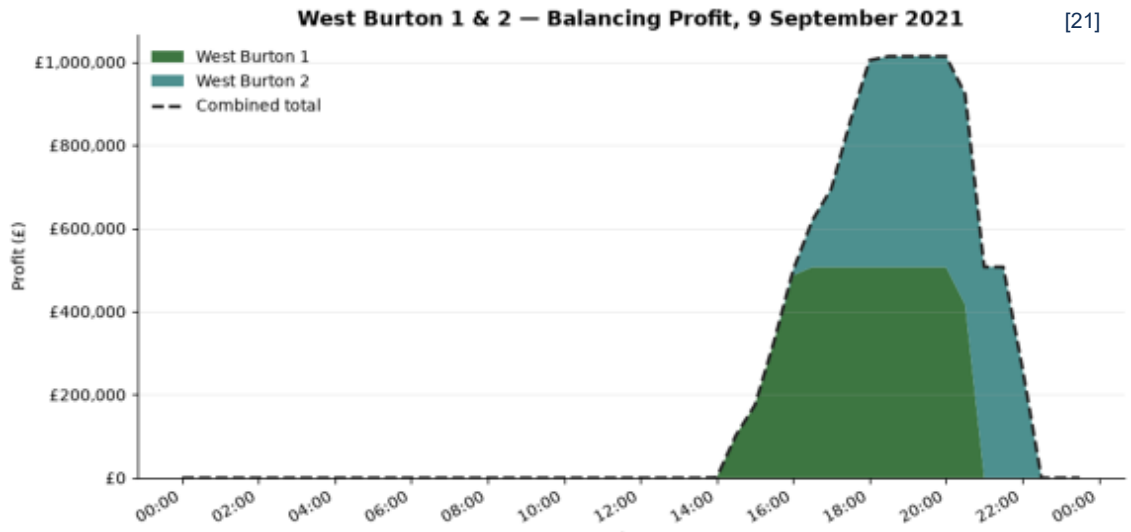
Type	Unit	Fuel	Price (£)	Volume (MW)	Flag	ID	Accepted
O	T_WBUPS-1	Coal	4,000	255		1	16:03
O	T_WBUPS-2	Coal	3,999	260		1	16:04
O	T_RYHPS-1	CCGT	3,700	270		1	16:02
O	T_SHBA-1	CCGT	3,700	400		1	16:04
O	T_CNQPS-4	CCGT	3,700	230		1	16:03
B	V_GHABI001	Battery	500	-33		-1	16:55
O	E_PETEM1	Gas Recip	455	230	⚑	1	16:39
B	HNOR-01	Battery	227.2	-4		-1	16:52
B	T_DINO-3	Pumped Storage	150	-300	⚑	-1	16:57
B	T_DINO-5	Pumped Storage	149	-117.5		-1	17:28
B	T_DINO-4	Pumped Storage	149	-242.5		-1	17:00
B	T_DINO-6	Pumped Storage	148	-102.5		-1	17:26
B	T_DAMC-1	CCGT	106	-173.95		-1	16:42
B	T_RATS-3	Coal	105.65	-4.17		-1	17:23
B	T_GRAI-8	CCGT	105.65	-43.75		-1	17:07

Period 37 accepted bids/offers – West Burton 1 & 2 among the top, unflagged [21]

£10.5m profit in 8 hours – EDF's combined West Burton 1 & 2 balancing-market profit on 9 September (£10,539,538): a quick and strategic bet on the system price that paid off enormously [21]



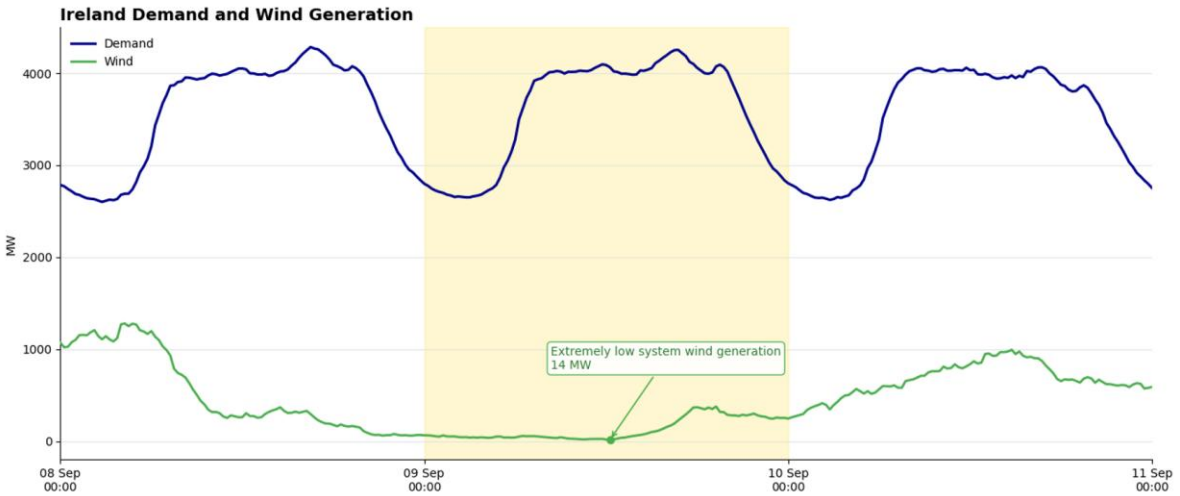
[21]



[21]

The Irish Side

- The same high-pressure system that stalled GB's wind turbines did the same across Ireland – wind generation fell to just **14 MW** [6]
- Several key units were out exactly when the system needed them most: Great Island (464 MW) returned from a planned outage at 15:00, then tripped twice more that evening on separate turbine faults [19]
- Whitegate, Huntstown 2, and both Tarbert oil units were already offline on extended maintenance – **removing 1.3 GW** of flexible capacity before the day even started
- Demand held around **4 GW** for most of the day and the system was short – until wind picked up and flipped it long right as the evening peak hit



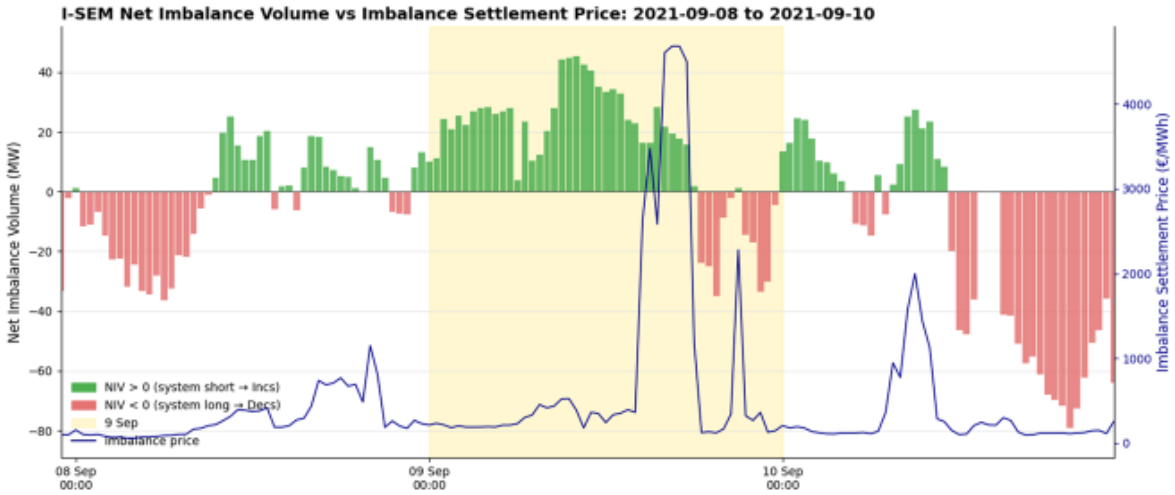
Unit	Unavailable (MW)	Fuel	Status on 9 Sept 2021
Great Island	464	Gas (CCGT)	Planned outage until 15:00, back briefly, then two separate HRSG faults that evening (19:00-20:20 and 22:30-02:30) – tripped twice right through the peak
Whitegate	444	Gas (CCGT)	Long-term outage since Dec 2020, still in extended testing
Huntstown 2	400	Gas (CCGT)	Out since late Jan 2021, testing still incomplete
Tarbert TB4	243	Oil	Turbine fault, out since April 2021 (long-term)
Tarbert TB3	241	Oil	Planned outage, 9 Aug - 19 Sept – meaning both Tarbert oil units were down simultaneously
Poolbeg PBA	232	Gas (CCGT)	Scheduled maintenance, extended
Poolbeg PBB	232	Gas (CCGT)	High exhaust temps – both Poolbeg units down, whole station offline
West Offaly	137	Peat	Permanent plant closure (not a day-specific outage – background capacity loss)
Lough Ree	91	Peat	Permanent plant closure (same caveat)
Aghada AT12	90	Gas	Maintenance, 4 Aug - 24 Sept
Aghada AT14	90	Gas	Short outage, 6-11 Sept
Moneypoint Unit 2	85	Coal	Partial derate (booster fan issue), available down to 200 MW
Tarbert TB1	54	Oil	Boiler fault – but this one came back online at 14:00 on the 9th, just as things were getting tight
Dublin Waste to Energy	22	Waste	Minor partial derate
Total Unavailable:			**2,825 MW**

Source: Nord Pool UMM Outage Portal [19]

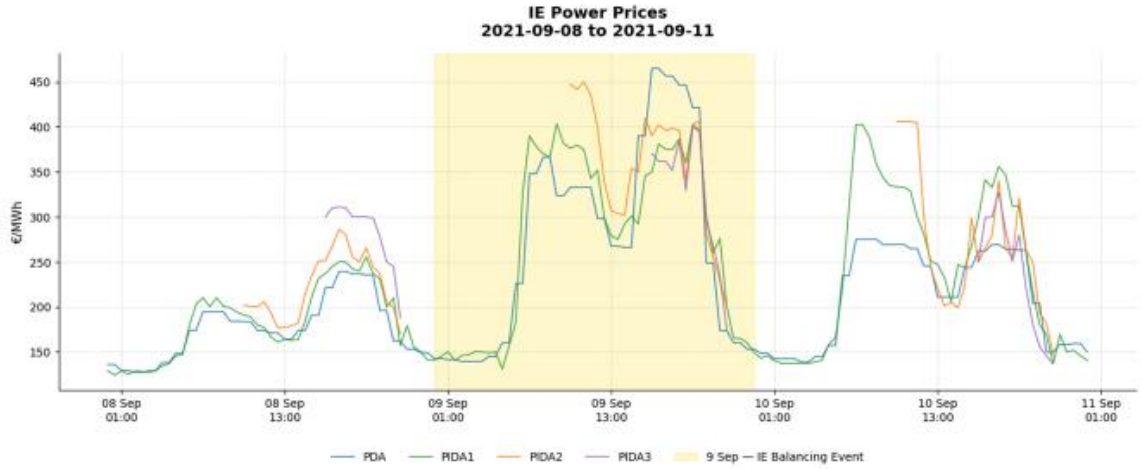
The Irish Side

How the GB price spike crossed the interconnectors into the I-SEM imbalance price

- Irish ex-ante prices track GB's clearing price – GB's EPEX DA auction settles at 9:20 GMT, before Ireland's SEMOpx auction at 11:00 – so when GB is tight, Ireland's DA price moves with it [12]
- Through 2021, GB had low plant margins and high prices, so the day-ahead spread often favoured Ireland exporting – even during periods when Ireland's own system was very tight
- SEMO's September 2021 report put average **Irish DA prices up 153%** (€37.59 → €95.13/MWh) on low wind, multiple plant outages and rising fuel costs [22]
- On 9 September, between 16:00 and 18:00, imbalance prices spiked to €4,680/MWh – with a record daily average price of €862.14/MWh [6], [22]
- Yet it wasn't Irish scarcity driving this: these extreme prices were set by the GB market, via SO-SO trades between the TSOs across the interconnectors [23]



I-SEM Net Imbalance Volume and Imbalance Settlement Price, 8–10 Sep 2021 [6]



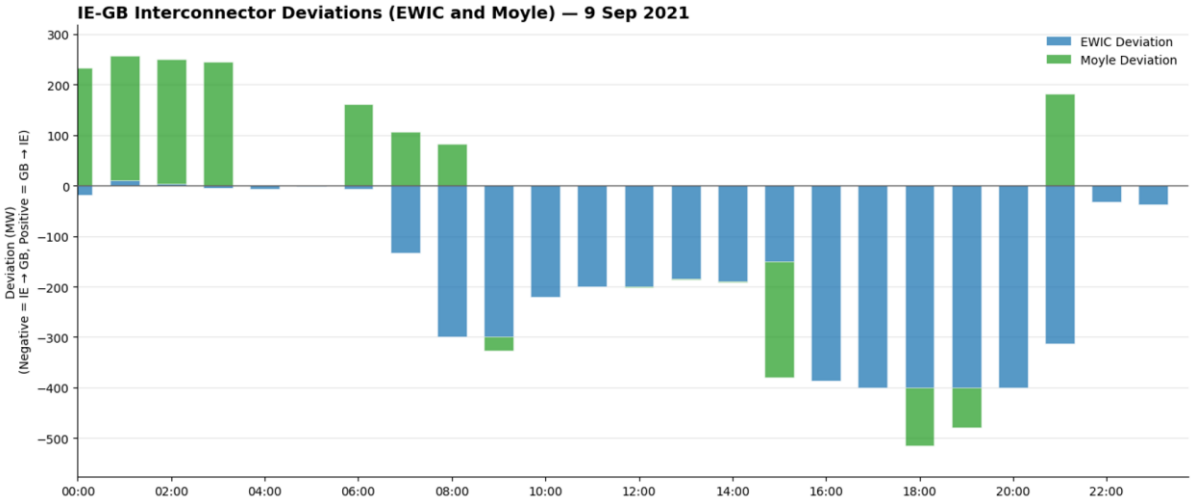
IE ex-ante power prices (DA, IDA1/2/3), 8–11 Sep 2021 [6]

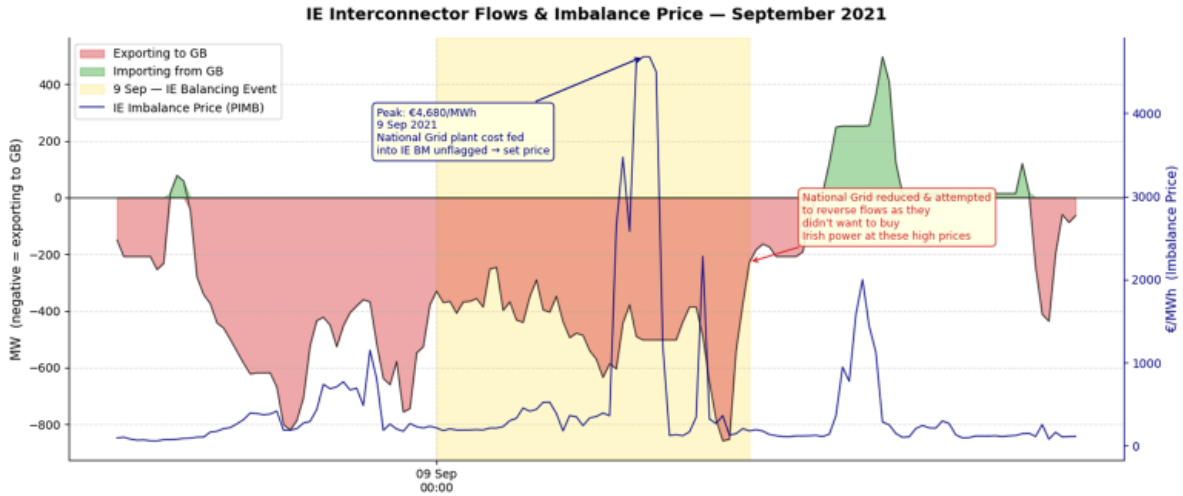
€862.14/MWh record daily-average Irish imbalance price on 9 Sept 2021 – set by GB-priced interconnector trades, not domestic scarcity [6]

TSO Balancing Interaction

How EirGrid/SONI's real-time SO-SO trades with National Grid set the Irish imbalance price

- On September 9th 2021 DA interconnector flows were scheduled to export power from Ireland to GB – however as seen on the chart to the right, these flows deviated massively. [6]
- As both systems tightened, EirGrid and SONI used SO-SO trades with National Grid to protect Ireland's security of supply – clawing back scheduled exports or pulling in extra imports [23]
- National Grid priced these trades off its own system price, so they were exceptionally high [22]
- Unflagged, these interconnector trades fed directly into the Irish balancing market – counting toward the imbalance price [22]
- The result: GB's system tightness was reflected indirectly in Ireland's own imbalance price by an old coal plant in England



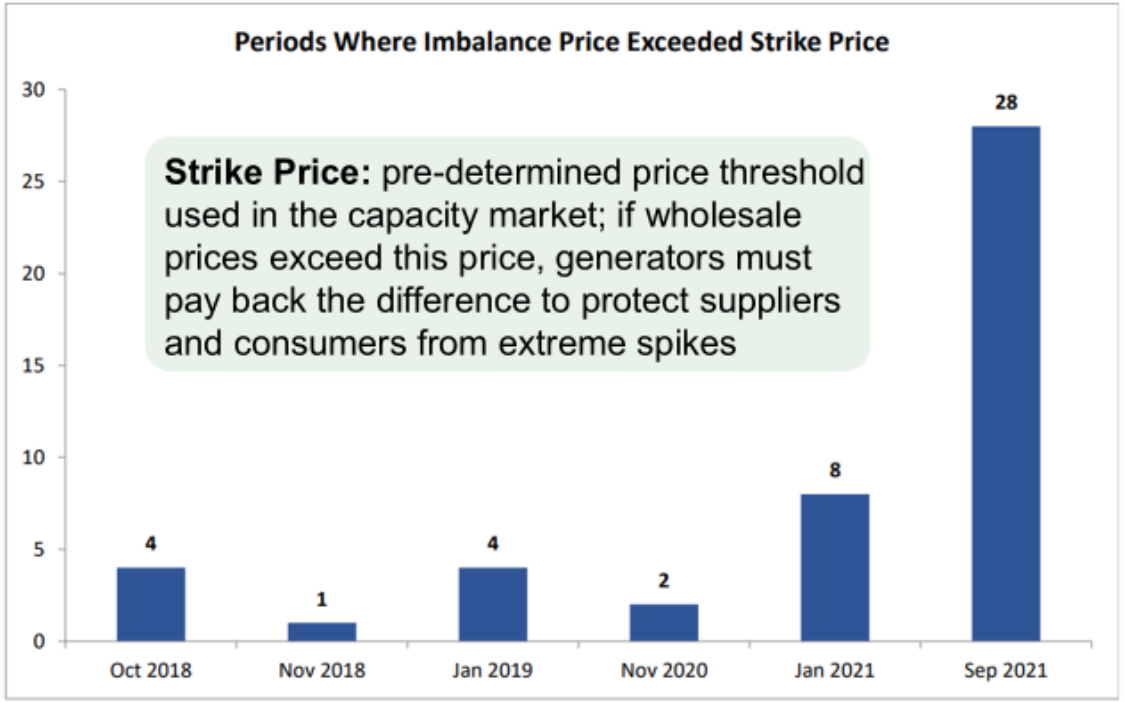


€4,680/MWh National Grid's SO-SO trade price fed into the Irish balancing market unflagged, and was used to set the imbalance price [6], [24]

Policy Changes and Learning Outcomes

Introduction of Mod_02_21 by SEMO

- Three modifications were raised in February 2021 to Appendix N of the Trading & Settlement Code (the flagging/tagging rules), targeting cases where SO-flaggable actions were setting the price instead [24]
- SEMO approved **Mod_02_21** – "Setting a Flag for Interconnector Auctions" – on 12 August 2021, to stop SO trade prices from setting the BM price [24]
- The fix **flags interconnector trades out of the imbalance price calculation**, recognising them as system-security actions, not genuine energy scarcity signals
- IT upgrades meant the fix couldn't go live until spring 2022 – leaving the second half of 2021 exposed, as seen on 9 September [24]
- According to SEMO analysis, 100% of Reliability Option (RO) strike-price breaches (12 Aug – 29 Sep 2021) were caused by SO trades that could have been flagged [22]



RO strike-price breaches by period, Oct 2018 – Sep 2021 (SEMO) [22]

However, modifying market policy only patches the problem – the underlying issue is clear: **The Irish grid needs more flexible generation**



Ireland's Need for More Flexible Generation

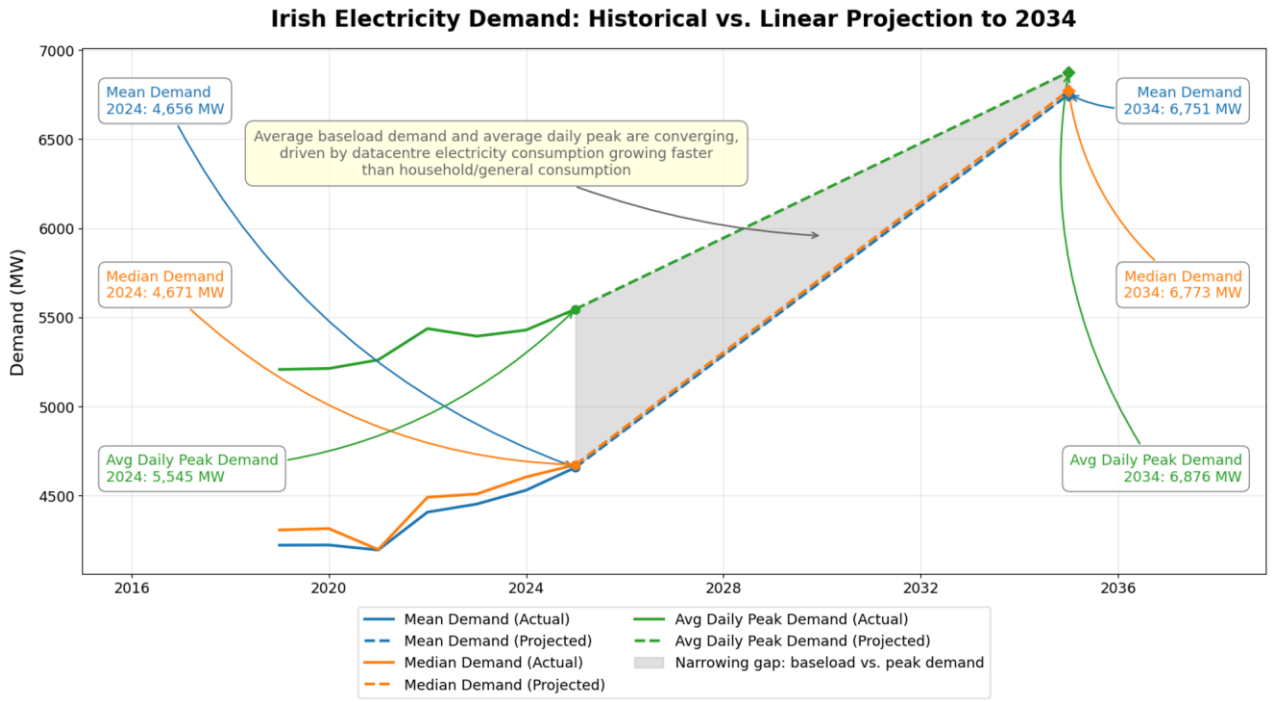
Why Ireland Needs More Flexible Capacity

Demand Side

- Electricity **demand is set to grow by 45%** from its 2023 to 2034, with the **daily peak set to rise by 24%.** [25]
- **Data centres** used 23% of metered generation in 2025 (7,633 GWh) – an increase of 10% from 2024. Household demand rose by just 2% over the same period. [26]
- Data centres and new technologies are set to reach **31% of demand** by 2034. [25]

Supply Side

- **Moneypoint stopped** burning coal in June 2025, meaning Ireland **lost 915 MW of firm capacity** – it remains as an oil-fired emergency backup until 2029. [27]
- EirGrid has described a “challenging outlook” until 2027, with temporary emergency generation held in reserve in case of shortfalls. [25]
- Approximately **649 MW of temporary emergency generation** now **sits outside the market** as a last resort backup. [28]



Extrapolation of Irish electricity demand [6], [25]

The Key Issue:

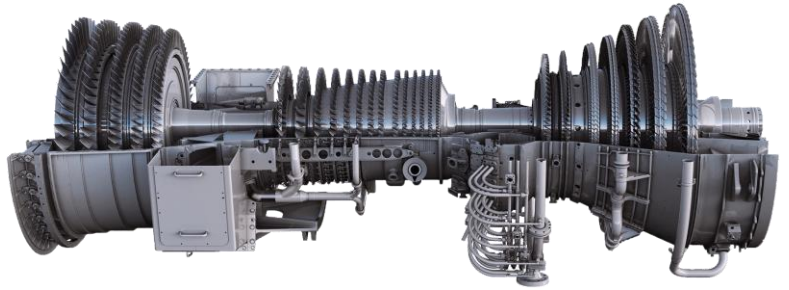
Demand is growing faster than firm supply is being built.

Every new GW of wind adds more intermittent variability, not certainty.

More gas and low carbon synchronous generation is needed to stabilise and secure the grid.

Building Low-Carbon Backup

Flexible generation: which machine, which fuel, and what is being built



WHICH MACHINE? [32], [33]

Heavy-duty OCGT

Start **~10 min**

Eff. **38 – 43%**

Cheapest per MW at scale

Aeroderivative OCGT

Start **~5 min**

Eff. **37 – 41%**

Fast, compact sites

Gas engines

Start **2 – 5 min**

Eff. **45 – 48%**

Frequent starts, efficient at part-load

WHICH FUEL? fossil → renewable

Natural gas	~2 GW new flexible gas by 2030 [5], [31]; emergency LNG reserve ~2030 [32]
HVO	Waste-oil renewable diesel; **~2× diesel price**, mostly imported [33]
Biomethane	5.7 TWh target by 2030, via gas grid [34]
Hydrogen	Flexible power an end use; 2030s+ [35]

CURRENT PROJECTS

Project	Size	Technology & fuel	Online
Tarbert Next Gen (SSE), Kerry [36]	**300 MW**	OCGT, designed for 100% HVO	End of 2027
Platin (SSE), Meath [37]	**180 MW**	OCGT, gas or biofuels; H_2-convertible	2028
Kilshane Ph. 1 & 2, Dublin [38]	**293 + 680 MW**	OCGT peakers	2028 / 2029
Silvermines pumped hydro, Tipperary [39]	**296 MW, ~7 h**	Long-duration storage	In permitting
4 new synchronous condensers [40]	**6,936 MVA**	Inertia, reactive power and frequency stabilisation	2027 / 2028

Emergency Gen: four temporary emergency fossil-fuel plants sit outside the market, and will cost homes and businesses **~€1.44bn** despite only ever being called on just twice (August solar eclipse and June 2026) [41]

Conclusions & Recommendations

WHAT WAS FOUND

1. **9 Sep 2021 was a warning.** Outages plus low wind left Ireland short of flexible capacity; the market regulation changes solved the price spikes, not the generation shortage.
2. **The gap is widening.** Demand +45% by 2034 [25], data centres already 23% of consumption [26], while firm plants like Moneypoint retire.
3. **Storage is necessary, not sufficient.** Batteries cover hours, storing energy generated by renewables; but the gaps can last days.

WHAT SHOULD HAPPEN

Build Low Carbon Flexible Generation

Deliver the 2 GW target [5] with units that burn HVO, biomethane or H_2 and run rarely

Add More Long-Term Storage

4–10 h+ projects like Silvermines [39], not just more short batteries

Regulate Data Centre Consumption

Enforce the CRU policy: on-site capacity must be dispatchable when the grid is tight [42]

More Interconnection and Stability

Deliver Celtic IC (French nuclear and different weather to GB) [43] and synchronous condensers for inertia [40]

Conclusion: Market volatility is an economic reflection of gaps in the physical grid, and as demand grows, so will price spikes. To prevent these costly events from occurring, more flexible generation is needed.

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